Welcome to Girls Who Code

LESSON 0

September 13th, 2026

Agenda

01 What is Girls Who Code?

02 Ice Breaker

03 Woman in Tech

04 Lesson

05 Practice Exercises

06 Reflection

What is Girls Who Code?

Girls Who Code is a nonprofit organization which aims to support and increase the number of women in computer science by equipping young women with the necessary computing skills to pursue 21st century opportunities.

girls who code

What is Girls Who Code?

Let’s watch a video about **Girls Who Code…**

https://www.youtube.com/watch?v=KZkhuS5C8eQ

Ice Breaker #1 !

Find someone you've never met before and share the following:

1. Name
2. Grade
3. Learn three things about your partner

Ice Breaker #2!

1. We'll start with a category of your choosing.

2. Someone will come up with a related word.

3. Then, the next person in your group will come up with a word that start with the LAST LETTER of the word that the person before them said and so on.

4. If you take more than 20 seconds to answer, you are eliminated from the game until the next round!

WOMAN IN TECH: **Grace Hopper**



Reflection Questions

1. How did Grace Hopper change the ways we communicate with computers?
2. Name two jobs/careers/areas Grace Hopper worked in.
3. When/Where else can we use computer science?

https://www.youtube.com/watch?v=Fg82iV-L8ZY

What will we do this semester?

First Semester Key Topics:

- Python basics
- Artificial Intelligence
- Create your own pet project

First Semester	
Date	**Subject**
13-Sep	Setup day & Introduction to Variables
20-Sep	Variables, Types
27-Sep	Conditional Statements
4-Oct	Lists
11-Oct	Fall Break No Class
18-Oct	For Loops, Functions
25-Oct	Methods, Classes, constructors
1-Nov	Introduction Artificial Intelligence
8-Nov	Artificial Intelligence Continued
15-Nov	Project Work Day
22-Nov	Semester Wrap Up & Project Presentation Day!

5 MINUTE BREAK!



What is coding?

Computer Science: The study of how computers can be used to make things easier, faster, or more fun.

Coding: Writing instructions for a computer in a language the computer can understand.



Computer Scientist: A person who writes code for a computer to do a task. A computer scientist's job often includes being a programmer, designer, and problem solver.

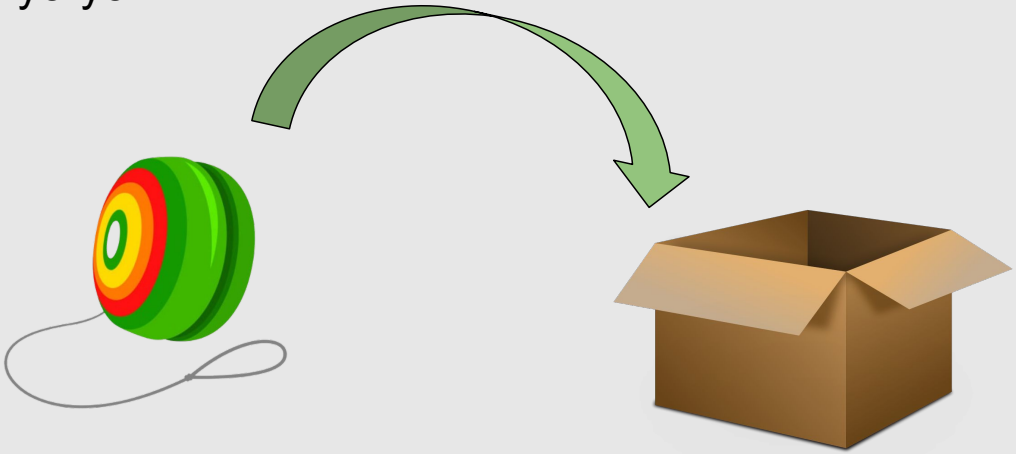
What is a Python?

Welcome to Python
(video link)



What is a Variable?

Think of variables as containers for storing data values…
Like a box for toys. One type of toy you could store would be a yo-yo.



So, what types of variables are there?

1. **String**
   Words, sentence, character, phrases
2. **Int**
   Numbers
3. **Boolean**
   True or False
4. **Float**
   Numbers with decimal places

```
String
country = “United States”
country = ‘United States’

int
year_founded = 1776

Boolean
is_big = True

Float
USD_to_EUR = 0.937
```

Creating Variables

In Python, there is no command to create a variable, so it's created the moment you declare it

The type doesn't need to be declared, so it can change at any time.

Variables are case-sensitive and can only contain numbers, letters, or underscores.

They start with underscores or letters.

Strings can be declared with single or double quotes

```
my age = 18 wrong
1my_age = 18 wrong
my_age = 18 right
My_age does NOT equal my_age
```

Why use variables?

If you wanted to print “Girls Who Code” many times, you wouldn’t want to type it out each time. What if we rebranded to Emory Girls Who Code? You would have to change everything.

You could create this string:

What would this print?

```
name = “Girls Who Code”
print(name)
print(name)
print(name)
```

Why use variables?

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You could create this string:

What would this print?

```
name = “Girls Who Code”
print(name)
print(name)
print(name)
```

```
Output →
Girls Who Code
Girls Who Code
Girls Who Code
```

Example

```
k = "helloooo"
print(k)
k = 15
print(k)
k = 'f'
print(k)
```

Can you tell which data types k becomes in each new assignment?

What's the output?

Example

```
k = "helloooo"
print(k)
k = 15
print(k)
k = 'f'
print(k)
```

Can you tell which data types k becomes in each new assignment?

What's the output?

```
Output →
helloooo
15
f
```

Activity in TextJam

https://tinyurl.com/yht73bzv

What will we do next class?

More information about variables + types!

THANK YOU!

